DBMS PROJECT REPORT

CRICKET PLAYER PERFORMANCE & SELECTION INTELLIGENCE SYSTEM

DBMS Experiential Learning – Level 3 Project

Submitted by

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1. PROJECT TITLE

Cricket Player Performance & Selection Intelligence System

1.1 Problem Statement

Cricket teams and selection committees need to evaluate players based on their performance across different match formats, venues, opponents and playing conditions. Traditional evaluation based only on overall career statistics may not provide sufficient information about a player's performance in specific situations.

There is a need for a centralized database system that stores detailed cricket statistics and enables efficient analysis of individual player performances. This project addresses this problem by developing a database-driven system that helps organize, retrieve and analyze cricket data to support informed player selection decisions.

1.2 Objectives

The main objectives of the project are:

1. To design and implement a relational database for storing cricket-related information such as players, teams, matches, venues and performance records.
2. To maintain detailed batting, bowling and dismissal statistics for individual players.
3. To analyze player performance based on different match formats, opponents, venues and match conditions.
4. To implement SQL operations such as joins, subqueries, views, stored procedures, triggers and aggregate queries.
5. To use MongoDB for storing flexible scouting reports and player observations.
6. To develop a web-based application using Next.js to display player information and performance analytics.
7. To support data-driven cricket player evaluation and selection decisions.

1.3 Scope of the Project

The project focuses on organizing and analyzing men's international cricket data, particularly One Day International (ODI) and Twenty20 International (T20I) formats.

The system covers the following areas:

- Player profiles and playing roles.
- Team and opponent information.
- Match and innings details.

- Venue and weather information.
- Batting and bowling performance.
- Player dismissal records.
- Player performance analysis and selection support.
- Scouting reports stored using MongoDB.

The application provides a centralized platform for managing cricket statistics and generating meaningful performance reports. It is designed as a decision-support system for player evaluation and selection rather than a live-scoring or betting application.

2. STUDENT SRNs AND NAMES

The Cricket Player Performance & Selection Intelligence System is developed as a team project by the following students.

Sl. No.	Student Name	SRN	Project Contribution
1	P.Jaya Surya	PES1UG24AM210	SQL database design, schema development and backend API implementation
2	P.S.Sai Sreyaesh	PES1UG24AM189	Next.js frontend development and work related to the proposed vector database component

Both team members contribute to the development of the system, including database design, application development, integration and project documentation.

3. ABSTRACT

The Cricket Player Performance & Selection Intelligence System is a database-driven web application developed to organize, manage and analyze cricket player statistics. The main purpose of this project is to support data-driven player evaluation and selection by providing detailed information about individual player performances across different match formats, opponents, venues and playing conditions.

The system uses MySQL as its primary relational database to store structured cricket information, including player details, teams, matches, innings, venues, weather conditions, batting performances, bowling performances, dismissals and player roles. The database is designed using primary keys, foreign keys and integrity constraints to maintain data consistency and establish relationships between different entities.

MongoDB is used as a document-oriented NoSQL database to store flexible scouting reports and player observations. This provides flexibility in managing information that may vary between players and scouting assessments.

The web application is developed using Next.js, React and TypeScript. It provides an interface for accessing player information, viewing performance statistics and generating analytical reports. Backend API routes connect the application to the databases and handle data retrieval and application operations. Authentication and role-based authorization are also incorporated to protect restricted application functionality.

The system demonstrates important DBMS concepts, including data definition and manipulation, SQL queries, joins, aggregate functions, subqueries, views, stored procedures, triggers, indexes and transactions.

Overall, the project aims to provide a centralized platform for organizing cricket statistics and supporting informed player evaluation and selection decisions through database-driven analysis.

4. TECHNOLOGY STACK

The Cricket Player Performance & Selection Intelligence System uses a combination of frontend, backend and database technologies to develop a database-driven web application. The selected technologies help manage structured cricket statistics, store flexible scouting information and present analytical results through a web interface.

4.1 Frontend Technologies

Next.js: A React-based web framework used to develop the application interface, pages and API routes.

React: A JavaScript library used to build reusable and interactive user interface components.

TypeScript: A programming language that adds static typing to JavaScript and improves code reliability and maintainability.

Tailwind CSS: A utility-first CSS framework used to style the application's user interface.

4.2 Backend Technologies

Node.js: A JavaScript runtime used to execute server-side application code.

Next.js API Routes: Used to handle application requests, perform backend operations and communicate with the databases.

mysql2/promise: A Node.js MySQL driver used to establish database connections and execute SQL queries asynchronously.

MongoDB Node.js Driver: Used to connect the application to MongoDB and perform document-based database operations.

JWT (JSON Web Token): Used to create and verify authentication sessions.

bcrypt: Used for secure password hashing and password verification.

4.3 Database Technologies

MySQL:

MySQL is the primary relational database management system used to store structured cricket information. It manages player details, teams, matches, innings, venues, weather conditions and player performance records. It also supports relational operations such as joins, constraints, views, stored procedures and triggers.

MongoDB:

MongoDB is a document-oriented NoSQL database used to store flexible scouting reports and player observations. It allows information to be stored in JSON-like documents without requiring every document to have an identical structure.

4.4 Development Tools

Visual Studio Code: Used for writing, editing and managing the application source code.

MySQL Workbench: Used for database creation, SQL query execution, schema inspection and database management.

MongoDB Shell (mongosh): Used to execute MongoDB commands and inspect document collections.

Git and GitHub: Used for source-code version control and project repository management.

4.5 Database Information

Database System	Database Name	Purpose
MySQL	cricket_selection	Stores structured cricket and application data.
MongoDB	cricket_selection_nosql	Stores flexible scouting reports.

The MySQL database currently contains 12 tables and 2 views, making a total of 14 confirmed database objects.

4.6 Application Architecture

The application follows a client-server architecture in which the frontend communicates with backend API routes. These API routes perform database operations using the appropriate database drivers.

The application follows this data flow:

1. The user interacts with the Next.js frontend.
2. The frontend sends HTTP requests to the backend API routes.
3. The API routes perform authentication and authorization checks where required.
4. The backend communicates with MySQL or MongoDB using their respective database drivers.
5. The database returns the requested information.
6. The API sends the response back to the frontend for display.

This architecture separates the user interface, backend operations and database management, making the application easier to maintain and extend.

5. SQL – DATABASE COMMANDS

MySQL is used as the primary relational database management system in the Cricket Player Performance & Selection Intelligence System. It stores structured information related to players, teams, matches, innings, venues, weather conditions and player performance.

This section demonstrates the SQL commands used to create, inspect, manipulate and retrieve information from the project database. It also covers database constraints, joins, aggregate functions, subqueries, views, stored procedures, triggers, indexes and transactions.

5.1 Database Selection and Table Inspection

The project uses a MySQL database named cricket_selection. The following commands are used to select the database and inspect its tables and structure.

5.1 Database selection and inspection

```
mysql> USE cricket_selection;
Database changed
mysql>
mysql> SHOW TABLES;
+----------------------------+
| Tables_in_cricket_selection |
+----------------------------+
| batting_performance        |
| bowling_performance        |
| dismissal                  |
| innings                    |
| match                      |
| opponent                   |
| player                     |
| player_role                |
| team                       |
| users                      |
| venue                      |
| vw_batting_performance     |
| vw_player_details          |
| weather                    |
+----------------------------+
14 rows in set (0.01 sec)
```

5.2 Inspect all 12 table structures

```
mysql>
mysql> SELECT DATABASE();DESC player;
+--------------------+
| DATABASE()         |
+--------------------+
| cricket_selection  |
+--------------------+
1 row in set (0.00 sec)

+---------------+--------------+------+-----+---------+----------------+
| Field         | Type         | Null | Key | Default | Extra          |
+---------------+--------------+------+-----+---------+----------------+
| player_id     | int          | NO   | PRI | NULL    | auto_increment |
| player_name   | varchar(100) | NO   |     | NULL    |                |
| country       | varchar(50)  | NO   |     | NULL    |                |
| batting_style | varchar(30)  | NO   |     | NULL    |                |
| bowling_style | varchar(50)  | YES  |     | NULL    |                |
| date_of_birth | date         | YES  |     | NULL    |                |
+---------------+--------------+------+-----+---------+----------------+
6 rows in set (0.01 sec)

mysql>
mysql> DESC team;
+-----------+-------------+------+-----+---------+----------------+
| Field     | Type        | Null | Key | Default | Extra          |
+-----------+-------------+------+-----+---------+----------------+
| team_id   | int         | NO   | PRI | NULL    | auto_increment |
| team_name | varchar(50) | NO   | UNI | NULL    |                |
+-----------+-------------+------+-----+---------+----------------+
2 rows in set (0.04 sec)

mysql>
mysql> DESC `match`;
+----------------+-------------+------+-----+---------+----------------+
| Field          | Type        | Null | Key | Default | Extra          |
+----------------+-------------+------+-----+---------+----------------+
| match_id       | int         | NO   | PRI | NULL    | auto_increment |
| team1_id       | int         | NO   | MUL | NULL    |                |
| team2_id       | int         | NO   | MUL | NULL    |                |
| venue_id       | int         | NO   | MUL | NULL    |                |
| weather_id     | int         | YES  | MUL | NULL    |                |
| match_format   | varchar(10) | NO   |     | NULL    |                |
| match_date     | date        | NO   |     | NULL    |                |
| winner_team_id | int         | YES  | MUL | NULL    |                |
+----------------+-------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)

mysql>
mysql> DESC innings;
+-----------------+--------------+------+-----+---------+----------------+
| Field           | Type         | Null | Key | Default | Extra          |
+-----------------+--------------+------+-----+---------+----------------+
| innings_id      | int          | NO   | PRI | NULL    | auto_increment |
| match_id        | int          | NO   | MUL | NULL    |                |
| innings_number  | int          | NO   |     | NULL    |                |
| batting_team_id | int          | NO   | MUL | NULL    |                |
| bowling_team_id | int          | NO   | MUL | NULL    |                |
| total_runs      | int          | NO   |     | 0       |                |
| total_wickets   | int          | NO   |     | 0       |                |
| overs_played    | decimal(4,1) | YES  |     | NULL    |                |
+-----------------+--------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

```
mysql>
mysql> DESC weather;
+--------------------+---------------+------+-----+---------+----------------+
| Field              | Type          | Null | Key | Default | Extra          |
+--------------------+---------------+------+-----+---------+----------------+
| weather_id         | int           | NO   | PRI | NULL    | auto_increment |
| weather_condition  | varchar(50)   | NO   |     | NULL    |                |
| temperature_c      | decimal(5,2)  | YES  |     | NULL    |                |
| humidity_percent   | decimal(5,2)  | YES  |     | NULL    |                |
| wind_speed_kmh     | decimal(5,2)  | YES  |     | NULL    |                |
| rainfall_mm        | decimal(6,2)  | YES  |     | NULL    |                |
+--------------------+---------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)

mysql>
mysql> DESC opponent;
+-------------+------+------+-----+---------+----------------+
| Field       | Type | Null | Key | Default | Extra          |
+-------------+------+------+-----+---------+----------------+
| opponent_id | int  | NO   | PRI | NULL    | auto_increment |
| player_id   | int  | NO   | MUL | NULL    |                |
| team_id     | int  | NO   | MUL | NULL    |                |
| match_id    | int  | NO   | MUL | NULL    |                |
+-------------+------+------+-----+---------+----------------+
4 rows in set (0.01 sec)

mysql>
mysql> DESC player_role;
+-----------+-------------+------+-----+---------+----------------+
| Field     | Type        | Null | Key | Default | Extra          |
+-----------+-------------+------+-----+---------+----------------+
| role_id   | int         | NO   | PRI | NULL    | auto_increment |
| player_id | int         | NO   | MUL | NULL    |                |
| format    | varchar(10) | NO   |     | NULL    |                |
| role      | varchar(50) | NO   |     | NULL    |                |
+-----------+-------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)
```

```
mysql>
mysql> DESC batting_performance;
+------------------+---------------+------+-----+---------+----------------+
| Field            | Type          | Null | Key | Default | Extra          |
+------------------+---------------+------+-----+---------+----------------+
| batting_perf_id  | int           | NO   | PRI | NULL    | auto_increment |
| innings_id       | int           | NO   | MUL | NULL    |                |
| player_id        | int           | NO   | MUL | NULL    |                |
| runs_scored      | int           | NO   |     | 0       |                |
| balls_faced      | int           | NO   |     | 0       |                |
| fours            | int           | NO   |     | 0       |                |
| sixes            | int           | NO   |     | 0       |                |
| strike_rate      | decimal(6,2)  | YES  |     | NULL    |                |
| batting_position | int           | YES  |     | NULL    |                |
| phase            | varchar(20)   | YES  |     | NULL    |                |
+------------------+---------------+------+-----+---------+----------------+
10 rows in set (0.01 sec)

mysql>
mysql> DESC bowling_performance;
+-----------------+---------------+------+-----+---------+----------------+
| Field           | Type          | Null | Key | Default | Extra          |
+-----------------+---------------+------+-----+---------+----------------+
| bowling_perf_id | int           | NO   | PRI | NULL    | auto_increment |
| innings_id      | int           | NO   | MUL | NULL    |                |
| player_id       | int           | NO   | MUL | NULL    |                |
| overs_bowled    | decimal(4,1)  | NO   |     | 0.0     |                |
| runs_conceded   | int           | NO   |     | 0       |                |
| wickets_taken   | int           | NO   |     | 0       |                |
| economy_rate    | decimal(5,2)  | YES  |     | NULL    |                |
| phase           | varchar(20)   | YES  |     | NULL    |                |
+-----------------+---------------+------+-----+---------+----------------+
8 rows in set (0.01 sec)

mysql>
mysql> DESC dismissal;
+-----------------+-------------+------+-----+---------+----------------+
| Field           | Type        | Null | Key | Default | Extra          |
+-----------------+-------------+------+-----+---------+----------------+
| dismissal_id    | int         | NO   | PRI | NULL    | auto_increment |
| batting_perf_id | int         | YES  | MUL | NULL    |                |
| dismissal_type  | varchar(30) | NO   |     | NULL    |                |
| bowler_id       | int         | YES  | MUL | NULL    |                |
| fielder_id      | int         | YES  | MUL | NULL    |                |
+-----------------+-------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)

mysql>
mysql> DESC users;
+---------------+-----------------------+------+-----+-------------------+-------------------+
| Field         | Type                  | Null | Key | Default           | Extra             |
+---------------+-----------------------+------+-----+-------------------+-------------------+
| user_id       | int                   | NO   | PRI | NULL              | auto_increment    |
| name          | varchar(100)          | NO   |     | NULL              |                   |
| email         | varchar(150)          | NO   | UNI | NULL              |                   |
| password_hash | varchar(255)          | NO   |     | NULL              |                   |
| role          | enum('admin','analyst') | NO   |     | analyst           |                   |
| created_at    | timestamp             | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+---------------+-----------------------+------+-----+-------------------+-------------------+
6 rows in set (0.03 sec)
```

5.3 Inspect table definitions and constraints

```
mysql> SHOW CREATE TABLE player;
+--------+-------------------
```

```
mysql>
mysql> SHOW CREATE TABLE innings;
+---------+-----------------------
```

```
mysql>
mysql> SHOW CREATE TABLE weather;
+---------+-----------------------
```

```
mysql>
mysql> SHOW CREATE TABLE opponent;
+----------+-----------------------
```

```
mysql>
mysql> SHOW CREATE TABLE batting_performance;
+---------------------+-----------------------
```

```
mysql>
mysql> SHOW CREATE TABLE dismissal;
+------------+-----------------------
```

```
mysql> CREATE TABLE demo_table (
    ->     id INT PRIMARY KEY,
    ->     name VARCHAR(100) NOT NULL
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql>
mysql> DESC demo_table;
+-------+--------------+------+-----+---------+-------+
| Field | Type         | Null | Key | Default | Extra |
+-------+--------------+------+-----+---------+-------+
| id    | int          | NO   | PRI | NULL    |       |
| name  | varchar(100) | NO   |     | NULL    |       |
+-------+--------------+------+-----+---------+-------+
2 rows in set (0.01 sec)

mysql>
mysql> ALTER TABLE demo_table
    -> ADD COLUMN country VARCHAR(50);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> DESC demo_table;
+---------+--------------+------+-----+---------+-------+
| Field   | Type         | Null | Key | Default | Extra |
+---------+--------------+------+-----+---------+-------+
| id      | int          | NO   | PRI | NULL    |       |
| name    | varchar(100) | NO   |     | NULL    |       |
| country | varchar(50)  | YES  |     | NULL    |       |
+---------+--------------+------+-----+---------+-------+
3 rows in set (0.00 sec)

mysql>
mysql> DROP TABLE demo_table;
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO player
    -> (player_name, country, batting_style)
    -> VALUES
    -> ('Demo Player 2026', 'India', 'Right hand bat');
Query OK, 1 row affected (0.01 sec)
```

5.5 DML – Data Manipulation Language

```
mysql> INSERT INTO player
    -> (player_name, country, batting_style)
    -> VALUES
    -> ('Demo Player 2026', 'India', 'Right hand bat');
Query OK, 1 row affected (0.01 sec)

mysql>
mysql> SELECT *
    -> FROM player
    -> WHERE player_name = 'Demo Player 2026';
+-----------+------------------+---------+----------------+---------------+---------------+
| player_id | player_name      | country | batting_style  | bowling_style | date_of_birth |
+-----------+------------------+---------+----------------+---------------+---------------+
|      8016 | Demo Player 2026 | India   | Right hand bat | NULL          | NULL          |
+-----------+------------------+---------+----------------+---------------+---------------+
1 row in set (0.02 sec)

mysql>
mysql> UPDATE player
    -> SET country = 'India'
    -> WHERE player_name = 'Demo Player 2026';
Query OK, 0 rows affected (0.01 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql>
mysql> DELETE FROM player
    -> WHERE player_name = 'Demo Player 2026';
Query OK, 1 row affected (0.01 sec)
```

5.6 SELECT, WHERE, DISTINCT, ORDER BY and LIMIT

```
mysql> SELECT *
    -> FROM player;
```

player_id	player_name	country	batting_style	bowling_style	date_of_birth
1	Virat Kohli	India	Right-handed	NULL	NULL
2	M Klinger	Australia	Right hand Bat		1980-07-04
3	TM Head	Australia	Left hand Bat	Right arm Offbreak	1993-12-29
4	MC Henriques	Australia	Right hand Bat	Right arm Fast medium	1987-02-01
5	AJ Turner	Australia	Right hand Bat	Right arm Offbreak	1993-01-25
6	JP Faulkner	Australia	Right hand Bat	Left arm Fast medium	1990-04-29
7	TD Paine	Australia	Right hand Bat	Right arm Medium	1984-12-08
8	PJ Cummins	Australia	Right hand Bat	Right arm Fast	1993-05-08
9	A Zampa	Australia	Right hand Bat	Legbreak Googly	1992-03-31
10	B Stanlake	Australia	Left hand Bat	Right arm Fast	1994-11-04
11	AJ Tye	Australia	Right hand Bat	Right arm Medium fast	1986-12-12
12	N Dickwella	Sri Lanka	Left hand Bat		1993-06-23
13	WU Tharanga	Sri Lanka	Left hand Bat		1985-02-02
14	EMDY Munaweera	Sri Lanka	Right hand Bat	Right arm Offbreak	1989-04-24
15	DAS Gunaratne	Sri Lanka	Right hand Bat	Right arm Medium fast	1986-01-08
16	TAM Siriwardana	Sri Lanka	Left hand Bat	Slow Left arm Orthodox	1985-12-04
17	CK Kapugedera	Sri Lanka	Right hand Bat	Right arm Medium	1987-02-24
18	S Prasanna	Sri Lanka	Right hand Bat	Legbreak	1985-06-27
19	JRMVB Sanjaya	Sri Lanka	Right hand Bat	Right arm Fast medium	1992-02-10
20	SL Malinga	Sri Lanka	Right hand Bat	Right arm Fast	1983-08-28
21	WMDN Kulasekara	Sri Lanka	Right hand Bat	Right arm Fast medium	1982-07-22
22	PADLR Sandakan	Sri Lanka	Right hand Bat	Left arm Wrist spin	1991-06-10
23	BR Dunk	Australia	Left hand Bat	Right arm Offbreak	1987-03-11
24	JA Richardson	Australia	Right hand Bat	Right arm Fast	1996-09-20
25	BKG Mendis	Sri Lanka	Right hand Bat	Legbreak	1995-02-02
26	MD Shanaka	Sri Lanka	Right hand Bat	Right arm Medium	1991-09-09
27	Nizakat Khan	Hong Kong	Right hand Bat	Legbreak	1992-07-08
28	Aizaz Khan	Hong Kong	Right hand Bat	Right arm Medium fast	1993-03-21
29	Babar Hayat	Hong Kong	Right hand Bat	Right arm Medium	1992-01-05
30	Anshuman Rath	Hong Kong	Left hand Bat	Slow Left arm Orthodox	1997-11-05
31	Ehsan Khan	Hong Kong	Right hand Bat	Right arm Offbreak	1984-12-27
32	Tanwir Afzal	Hong Kong	Right hand Bat	Right arm Fast medium	1988-06-12
33	Shahid Wasif	Hong Kong	Right hand Bat	Right arm Offbreak	1997-05-15
34	Tanveer Ahmed	Hong Kong	Right hand Bat	Right arm Medium fast	1997-09-18
35	Ehsan Nawaz	Hong Kong	Right hand Bat	Right arm Fast medium	1995-03-21
36	Nadeem Ahmed	Hong Kong	Right hand Bat	Slow Left arm Orthodox	1987-09-28
37	C Carter	Hong Kong	Right hand Bat		1997-09-09
38	WTS Porterfield	Ireland	Left hand Bat	Right arm Offbreak	1984-09-06
39	DA Rankin	Ireland	Right hand Bat	Right arm Offbreak	1987-08-02
40	SP Terry	Ireland	Right hand Bat	Right arm Offbreak	1991-08-01
41	JNK Shannon	Ireland	Right hand Bat	Right arm Offbreak	1990-02-12
42	WJ O'Brien	Ireland	Right hand Bat	Right arm Medium fast	1984-03-04
43	GJ Thompson	Ireland	Right hand Bat	Legbreak	1987-09-17
44	AR McBrine	Ireland	Left hand Bat	Right arm Offbreak	1993-04-30
45	L Tucker	Ireland	Right hand Bat		1996-09-10
46	J Mulder	Ireland	Right hand Bat	Legbreak	1995-08-11
47	J Little	Ireland	Right hand Bat	Left arm Fast medium	1999-11-01
48	CA Young	Ireland	Right hand Bat	Right arm Fast medium	1990-04-04
49	KL Rahul	India	Right hand Bat		1992-04-18
50	Mandeep Singh	India	Right hand Bat	Right arm Medium	1991-12-18

```
mysql>
mysql> SELECT player_id, player_name, country
    -> FROM player;
```

player_id	player_name	country
1	Virat Kohli	India
2	M Klinger	Australia
3	TM Head	Australia
4	MC Henriques	Australia
5	AJ Turner	Australia
6	JP Faulkner	Australia
7	TD Paine	Australia
8	PJ Cummins	Australia
9	A Zampa	Australia
10	B Stanlake	Australia
11	AJ Tye	Australia
12	N Dickwella	Sri Lanka
13	WU Tharanga	Sri Lanka
14	EMDY Munaweera	Sri Lanka
15	DAS Gunaratne	Sri Lanka
16	TAM Siriwardana	Sri Lanka
17	CK Kapugedera	Sri Lanka
18	S Prasanna	Sri Lanka
19	JRMVB Sanjaya	Sri Lanka
20	SL Malinga	Sri Lanka
21	KMDN Kulasekara	Sri Lanka
22	PADLR Sandakan	Sri Lanka
23	BR Dunk	Australia
24	JA Richardson	Australia
25	BKG Mendis	Sri Lanka
26	MD Shanaka	Sri Lanka
27	Nizakat Khan	Hong Kong
28	Aizaz Khan	Hong Kong
29	Babar Hayat	Hong Kong
30	Anshuman Rath	Hong Kong
31	Ehsan Khan	Hong Kong
32	Tanwir Afzal	Hong Kong
33	Shahid Wasif	Hong Kong
34	Tanveer Ahmed	Hong Kong
35	Ehsan Nawaz	Hong Kong
36	Nadeem Ahmed	Hong Kong
37	C Carter	Hong Kong
38	WTS Porterfield	Ireland
39	DA Rankin	Ireland
40	SP Terry	Ireland
41	JNK Shannon	Ireland
42	KJ O'Brien	Ireland
43	GJ Thompson	Ireland
44	AR McBrine	Ireland
45	L Tucker	Ireland
46	J Mulder	Ireland
47	J Little	Ireland
48	CA Young	Ireland
49	KL Rahul	India
50	Mandeep Singh	India

```
014 rows in set (0.mysql>
mysql> SELECT player_name, country
    -> FROM player
    -> WHERE country = 'India';
+---------------------+---------+
| player_name         | country |
+---------------------+---------+
| Virat Kohli         | India   |
| KL Rahul            | India   |
| Mandeep Singh       | India   |
| AT Rayudu           | India   |
| MK Pandey           | India   |
| KM Jadhav           | India   |
| MS Dhoni            | India   |
| AR Patel            | India   |
| R Dhawan            | India   |
| JJ Bumrah           | India   |
| JD Unadkat          | India   |
| YS Chahal           | India   |
| DS Kulkarni         | India   |
| BB Sran             | India   |
| V Kohli             | India   |
| SK Raina            | India   |
| Yuvraj Singh        | India   |
| HH Pandya           | India   |
| Parvez Rasool       | India   |
| A Nehra             | India   |
| RR Pant             | India   |
| RG Sharma           | India   |
| AM Rahane           | India   |
| STR Binny           | India   |
| R Ashwin            | India   |
| RA Jadeja           | India   |
| Mohammed Shami      | India   |
| B Kumar             | India   |
| M Raj               | India   |
| S Mandhana          | India   |
| S Meghana           | India   |
| V Krishnamurthy     | India   |
| H Kaur              | India   |
| J Goswami           | India   |
| AA Patil            | India   |
| S Pandey            | India   |
| Preeti Bose         | India   |
| S Verma             | India   |
| E Bisht             | India   |
| A Poulose           | India   |
```

```
mysql>
mysql> uery aborted
mysql> SELECT DISTINCT country
    -> FROM player;
+--------------------------+
| country                  |
+--------------------------+
| India                    |
| Australia                |
| Sri Lanka                |
| Hong Kong                |
| Ireland                  |
| Zimbabwe                 |
| Bangladesh               |
| New Zealand              |
| South Africa             |
| England                  |
| West Indies              |
| United States of America |
| Pakistan                 |
| Scotland                 |
| Netherlands              |
| Oman                     |
| United Arab Emirates     |
| Afghanistan              |
| Papua New Guinea         |
| Nepal                    |
| Thailand                 |
| Uganda                   |
| Malaysia                 |
| Botswana                 |
| Lesotho                  |
| Malawi                   |
| Namibia                  |
| Sierra Leone             |
| Mozambique               |
| China                    |
| Kuwait                   |
| Philippines              |
| Vanuatu                  |
| Germany                  |
| Italy                    |
| Nigeria                  |
| Tanzania                 |
| Rwanda                   |
| Kenya                    |
| Japan                    |
| Indonesia                |
| Fiji                     |
| Samoa                    |
| Canada                   |
| Ghana                    |
| Guernsey                 |
| Denmark                  |
| Norway                   |
| Jersey                   |
| Maldives                 |
| Mali                     |
| Singapore                |
```

```
mysql>
mysql> SELECT player_name
    -> FROM player
    -> ORDER BY player_name ASC;
+-------------------------------+
| player_name                   |
+-------------------------------+
| A Adekunle                    |
| A Ahmadhel                    |
| A Aikenhead                   |
| A Aitken                      |
| A Aitken-Drummond             |
| A Akhigbe                     |
| A Akurut                      |
| A Alexander                   |
| A Alfonso                     |
| A Amado                       |
| A Amburose                    |
| A Amuzu                       |
| A Andrews                     |
| A Andriani                    |
| A Argyropoulou                |
| A Ashok                       |
| A Ashokan                     |
| A Athanaze                    |
| A Augastin                    |
| A Awasthi                     |
| A Bagai                       |
| A Balbirnie                   |
| A Beggs                       |
| A Beresford-Peirse            |
| A Bester                      |
| A Bhadauria                   |
| A Bhagwat                     |
| A Bhargava                    |
| A Bhattarai                   |
| A Bidari                      |
| A Bierwisch                   |
| A Bishnoi                     |
| A Bohara                      |
| A Bosch                       |
| A Bose                        |
| A Brindle                     |
| A Brown-Carrera               |
| A Bulow                       |
| A Bushell                     |
| A Busing-Volschenk            |
| A Canning                     |
| A Capsey                      |
| A Cespedes                    |
| A Chandel                     |
| A Charles                     |
| A Chaubey                     |
| A Chaudhary                   |
| A Chenthamara                 |
| A Cheung                      |
| A Chilia                      |
| A Chopra                      |
| A Codrington                  |
| A Conteh                      |
| A Cossa                       |
| A Curavic                     |
| A Dalzell                     |
| A Dananjaya                   |
| A Das                         |
| A Dastgir                     |
| A Davidson Soler              |
| A Davidson-Soler              |
| A Daxini                      |
| A Dcosta                      |
| A Denuwan                     |
| A Dercksen                    |
| A Desai                       |
| A Dhabe                       |
| A Dhar                        |
```

```
mysql>
mysql> SELECT *
    -> FROM batting_performance
    -> LIMIT 10;
+-----------------+-------------+-----------+-------------+-------------+-------+-------+-------------+------------------+-----------+
| batting_perf_id | innings_id  | player_id | runs_scored | balls_faced | fours | sixes | strike_rate | batting_position | phase     |
+-----------------+-------------+-----------+-------------+-------------+-------+-------+-------------+------------------+-----------+
|               1 |           1 |         1 |          43 |          34 |     2 |     2 |      126.47 |                1 | Middle    |
|               2 |           1 |         2 |          38 |          32 |     4 |     0 |      118.75 |                2 | Powerplay |
|               3 |           1 |         3 |          31 |          25 |     0 |     1 |      124.00 |                3 | Middle    |
|               4 |           1 |         4 |          17 |          10 |     0 |     1 |      170.00 |                4 | Middle    |
|               5 |           1 |         5 |          18 |          12 |     1 |     0 |      150.00 |                5 | Middle    |
|               6 |           1 |         6 |          14 |           7 |     2 |     0 |      200.00 |                6 | Death     |
|               7 |           2 |        12 |          30 |          25 |     4 |     1 |      120.00 |                1 | Powerplay |
|               8 |           2 |        13 |           0 |           2 |     0 |     0 |        0.00 |                2 | Powerplay |
|               9 |           2 |        14 |          44 |          29 |     6 |     0 |      151.72 |                3 | Powerplay |
|              10 |           2 |        15 |          52 |          37 |     7 |     0 |      140.54 |                4 | Middle    |
+-----------------+-------------+-----------+-------------+-------------+-------+-------+-------------+------------------+-----------+
10 rows in set (0.00 sec)
```

5.7 Aggregate functions

```
mysql>
mysql> SELECT player_name
    -> FROM player
    -> WHERE bowling_style IS NULL;
+-------------+
| player_name |
+-------------+
| Virat Kohli |
+-------------+
1 row in set (0.00 sec)
       SELECT COUNT(*) AS total_batting_records
    -> FROM batting_performance;
+------------------------+
| total_batting_records |
+------------------------+
|                 144473 |
+------------------------+
1 row in set (0.05 sec)

mysql>
mysql> SELECT SUM(runs_scored) AS total_runs
    -> FROM batting_performance;
+------------+
| total_runs |
+------------+
|    2597954 |
+------------+
1 row in set (0.25 sec)

mysql>
mysql> SELECT AVG(runs_scored) AS average_runs
    -> FROM batting_performance;
+--------------+
| average_runs |
+--------------+
|      17.9823 |
+--------------+
1 row in set (0.17 sec)

mysql>
mysql> SELECT MAX(runs_scored) AS highest_runs
    -> FROM batting_performance;
+--------------+
| highest_runs |
+--------------+
|          264 |
+--------------+
1 row in set (0.12 sec)

mysql>
mysql> SELECT MIN(runs_scored) AS lowest_runs
    -> FROM batting_performance;
+-------------+
| lowest_runs |
+-------------+
|           0 |
+-------------+
1 row in set (0.03 sec)
```

5.8 GROUP BY and HAVING

```
mysql> SELECT
    ->     p.player_id,
    ->     p.player_name,
    ->     SUM(bp.runs_scored) AS total_runs
    -> FROM player p
    -> JOIN batting_performance bp
    ->     ON p.player_id = bp.player_id
    -> GROUP BY p.player_id, p.player_name
    -> ORDER BY total_runs DESC;
```

player_id	player_name	total_runs
157	V Kohli	18788
165	RG Sharma	15574
6822	KC Sangakkara	13022
54	MS Dhoni	11858
110	AB de Villiers	11026
6821	TM Dilshan	10959
219	Babar Azam	10873
257	MJ Guptill	10642
258	LRPL Taylor	10100
108	Q de Kock	10000
259	DA Warner	9877
122	JC Buttler	9710
90	KS Williamson	9682
6820	DPMD Jayawardene	9531
121	EJG Morgan	9396
78	Tamim Iqbal	9272
107	HM Amla	9106
246	S Mandhana	8968
81	Shakib Al Hasan	8933
335	Mohammad Hafeez	8788
372	S Dhawan	8492
151	JE Root	8459
328	Mushfiqur Rahim	8403
220	Shoaib Malik	8373
140	CH Gayle	8365
1	Virat Kohli	8308
468	L Wolvaardt	8217
159	Yuvraj Singh	8068
382	SD Hope	7531
6756	MJ Clarke	7476
6758	RT Ponting	7380
83	Mahmudullah	7354
131	DA Miller	7351
6776	BB McCullum	7302
158	SK Raina	7134
25	BKG Mendis	7001
375	AD Mathews	6989
109	F du Plessis	6932
179	BL Mooney	6894
13	WU Tharanga	6889
208	AC Jayangani	6863
424	SR Taylor	6850
6846	SR Watson	6822
111	JP Duminy	6808
1173	SPD Smith	6758
307	PR Stirling	6708
249	H Kaur	6668
189	SW Bates	6596

```
mysql> SELECT
    ->     p.player_id,
    ->     p.player_name,
    ->     SUM(bp.runs_scored) AS total_runs
    -> FROM player p
    -> JOIN batting_performance bp
    ->     ON p.player_id = bp.player_id
    -> GROUP BY p.player_id, p.player_name
    -> HAVING SUM(bp.runs_scored) > 1000;
+-----------+----------------------+------------+
| player_id | player_name          | total_runs |
+-----------+----------------------+------------+
|         1 | Virat Kohli          |       8308 |
|         3 | TM Head              |       4283 |
|         6 | JP Faulkner          |       1184 |
|        12 | N Dickwella          |       2084 |
|        13 | WU Tharanga          |       6889 |
|        17 | CK Kapugedera        |       2180 |
|        25 | BKG Mendis           |       7001 |
|        21 | KMDN Kulasekara      |       1519 |
|        26 | MD Shanaka           |       3133 |
|        27 | Nizakat Khan         |       2896 |
|        28 | Aizaz Khan           |       1218 |
|        29 | Babar Hayat          |       2954 |
|        30 | Anshuman Rath        |       2700 |
|        38 | WTS Porterfield      |       3390 |
|        42 | KJ O'Brien           |       3561 |
|        45 | L Tucker             |       2420 |
|        60 | CJ Chibhabha         |       2499 |
|        61 | H Masakadza          |       5185 |
|        63 | Sikandar Raza        |       6392 |
|        64 | MN Waller            |       1546 |
|        66 | E Chigumbura         |       4035 |
|        49 | KL Rahul             |       5432 |
|        51 | AT Rayudu            |       1679 |
|        52 | MK Pandey            |       1267 |
|        53 | KM Jadhav            |       1440 |
|        54 | MS Dhoni             |      11858 |
|        55 | AR Patel             |       1656 |
|        75 | V Sibanda            |       2623 |
|        78 | Tamim Iqbal          |       9272 |
|        79 | Imrul Kayes          |       2361 |
|        80 | Sabbir Rahman        |       2198 |
|        81 | Shakib Al Hasan      |       8933 |
|        82 | Soumya Sarkar        |       3803 |
|        83 | Mahmudullah          |       7354 |
|        85 | Mashrafe Mortaza     |       1595 |
|        90 | KS Williamson        |       9682 |
|        91 | C Munro              |       2978 |
|        92 | CJ Anderson          |       1799 |
|        94 | C de Grandhomme      |       1247 |
|        95 | L Ronchi             |       1756 |
|       100 | JDS Neesham          |       2613 |
|        96 | MJ Santner           |       2728 |
|       107 | HM Amla              |       9106 |
|       108 | Q de Kock            |      10000 |
```

5.9 JOIN operations

```
mysql> SELECT p.player_name, bp.runs_scored
    -> FROM player p
    -> INNER JOIN batting_performance bp
    -> ON p.player_id = bp.player_id;
+--------------------------------+--------------+
| player_name                    | runs_scored  |
+--------------------------------+--------------+
| Virat Kohli                    |           43 |
| Virat Kohli                    |           12 |
| Virat Kohli                    |           53 |
| Virat Kohli                    |           20 |
| Virat Kohli                    |           36 |
| Virat Kohli                    |           18 |
| Virat Kohli                    |           42 |
| Virat Kohli                    |            8 |
| Virat Kohli                    |           84 |
| Virat Kohli                    |           68 |
| Virat Kohli                    |          172 |
| Virat Kohli                    |           16 |
| Virat Kohli                    |            3 |
| Virat Kohli                    |           47 |
| Virat Kohli                    |            7 |
| Virat Kohli                    |           27 |
| Virat Kohli                    |            0 |
| Virat Kohli                    |           28 |
| Virat Kohli                    |            0 |
| Virat Kohli                    |            3 |
| Virat Kohli                    |            1 |
| Virat Kohli                    |            1 |
| Virat Kohli                    |            0 |
| Virat Kohli                    |            8 |
| Virat Kohli                    |           64 |
| Virat Kohli                    |            0 |
| Virat Kohli                    |           37 |
| Virat Kohli                    |           37 |
| Virat Kohli                    |           17 |
| Virat Kohli                    |           52 |
| Virat Kohli                    |           42 |
| Virat Kohli                    |           14 |
| Virat Kohli                    |           55 |
| Virat Kohli                    |           46 |
| Virat Kohli                    |           40 |
| Virat Kohli                    |           39 |
| Virat Kohli                    |           35 |
| Virat Kohli                    |            0 |
| Virat Kohli                    |            1 |
| Virat Kohli                    |           12 |
| Virat Kohli                    |           69 |
| Virat Kohli                    |           79 |
| Virat Kohli                    |           36 |
| Virat Kohli                    |            4 |
| Virat Kohli                    |            6 |
| Virat Kohli                    |           30 |
| Virat Kohli                    |           53 |
```

```
mysql> SELECT p.player_name, bp.runs_scored
    -> FROM player p
    -> LEFT JOIN batting_performance bp
    -> ON p.player_id = bp.player_id;
+--------------------------------+--------------+
| player_name                    | runs_scored  |
+--------------------------------+--------------+
| Virat Kohli                    |           43 |
| Virat Kohli                    |           12 |
| Virat Kohli                    |           53 |
| Virat Kohli                    |           20 |
| Virat Kohli                    |           36 |
| Virat Kohli                    |           18 |
| Virat Kohli                    |           42 |
| Virat Kohli                    |            8 |
| Virat Kohli                    |           84 |
| Virat Kohli                    |           68 |
| Virat Kohli                    |          172 |
| Virat Kohli                    |           16 |
| Virat Kohli                    |            3 |
| Virat Kohli                    |           47 |
| Virat Kohli                    |            7 |
| Virat Kohli                    |           27 |
| Virat Kohli                    |            0 |
| Virat Kohli                    |           28 |
| Virat Kohli                    |            0 |
| Virat Kohli                    |            3 |
| Virat Kohli                    |            1 |
| Virat Kohli                    |            1 |
| Virat Kohli                    |            0 |
| Virat Kohli                    |            8 |
| Virat Kohli                    |           64 |
| Virat Kohli                    |            0 |
| Virat Kohli                    |           37 |
| Virat Kohli                    |           37 |
| Virat Kohli                    |           17 |
| Virat Kohli                    |           52 |
| Virat Kohli                    |           42 |
| Virat Kohli                    |           14 |
| Virat Kohli                    |           55 |
| Virat Kohli                    |           46 |
| Virat Kohli                    |           40 |
| Virat Kohli                    |           39 |
| Virat Kohli                    |           35 |
| Virat Kohli                    |            0 |
| Virat Kohli                    |            1 |
| Virat Kohli                    |           12 |
| Virat Kohli                    |           69 |
| Virat Kohli                    |           79 |
| Virat Kohli                    |           36 |
| Virat Kohli                    |            4 |
| Virat Kohli                    |            6 |
| Virat Kohli                    |           30 |
| Virat Kohli                    |           53 |
| Virat Kohli                    |           34 |
| Virat Kohli                    |            8 |
| Virat Kohli                    |           25 |
| Virat Kohli                    |           35 |
| Virat Kohli                    |            2 |
| Virat Kohli                    |            8 |
```

```
mysql> SELECT p.player_name, bp.runs_scored
    -> FROM player p
    -> RIGHT JOIN batting_performance bp
    -> ON p.player_id = bp.player_id;
```

player_name	runs_scored
Virat Kohli	43
M Klinger	38
TM Head	31
MC Henriques	17
AJ Turner	18
JP Faulkner	14
N Dickwella	30
WU Tharanga	0
EMDY Munaweera	44
DAS Gunaratne	52
TAM Siriwardana	15
CK Kapugedera	10
S Prasanna	8
M Klinger	43
Virat Kohli	12
BR Dunk	32
MC Henriques	56
TM Head	4
AJ Turner	7
JP Faulkner	1
TD Paine	7
PJ Cummins	3
AJ Tye	0
JA Richardson	0
N Dickwella	14
WU Tharanga	4
EMDY Munaweera	10
BKG Mendis	5
DAS Gunaratne	84
TAM Siriwardana	0
CK Kapugedera	32
S Prasanna	7
KMDN Kulasekara	12
SL Malinga	1
Virat Kohli	53
M Klinger	62
BR Dunk	28
TM Head	30
AJ Turner	1
MC Henriques	2
JP Faulkner	2
TD Paine	1
EMDY Munaweera	37
WU Tharanga	14
BKG Mendis	14

```
mysql> SELECT p.player_name, bp.runs_scored
    -> FROM player p
    -> LEFT JOIN batting_performance bp
    -> ON p.player_id = bp.player_id
    ->
    -> UNION
    ->
    -> SELECT p.player_name, bp.runs_scored
    -> FROM player p
    -> RIGHT JOIN batting_performance bp
    -> ON p.player_id = bp.player_id;
+-------------------------------+--------------+
| player_name                   | runs_scored  |
+-------------------------------+--------------+
| Virat Kohli                   |           43 |
| Virat Kohli                   |           12 |
| Virat Kohli                   |           53 |
| Virat Kohli                   |           20 |
| Virat Kohli                   |           36 |
| Virat Kohli                   |           18 |
| Virat Kohli                   |           42 |
| Virat Kohli                   |            8 |
| Virat Kohli                   |           84 |
| Virat Kohli                   |           68 |
| Virat Kohli                   |          172 |
| Virat Kohli                   |           16 |
| Virat Kohli                   |            3 |
| Virat Kohli                   |           47 |
| Virat Kohli                   |            7 |
| Virat Kohli                   |           27 |
| Virat Kohli                   |            0 |
| Virat Kohli                   |           28 |
| Virat Kohli                   |            1 |
| Virat Kohli                   |           64 |
| Virat Kohli                   |           37 |
| Virat Kohli                   |           17 |
| Virat Kohli                   |           52 |
| Virat Kohli                   |           14 |
| Virat Kohli                   |           55 |
| Virat Kohli                   |           46 |
| Virat Kohli                   |           40 |
| Virat Kohli                   |           39 |
| Virat Kohli                   |           35 |
| Virat Kohli                   |           69 |
| Virat Kohli                   |           79 |
| Virat Kohli                   |            4 |
| Virat Kohli                   |            6 |
| Virat Kohli                   |           30 |
| Virat Kohli                   |           34 |
| Virat Kohli                   |           25 |
| Virat Kohli                   |            2 |
| Virat Kohli                   |           44 |
| Virat Kohli                   |            9 |
| Virat Kohli                   |            5 |
| Virat Kohli                   |           13 |
| Virat Kohli                   |           31 |
| Virat Kohli                   |           63 |
| Virat Kohli                   |           61 |
| Virat Kohli                   |           24 |
| Virat Kohli                   |           29 |
| Virat Kohli                   |           58 |
```

5.10 Views

```
mysql> SHOW FULL TABLES
    -> WHERE Table_type = 'VIEW';
+-------------------------------+------------+
| Tables_in_cricket_selection | Table_type |
+-------------------------------+------------+
| vw_batting_performance      | VIEW       |
| vw_player_details           | VIEW       |
+-------------------------------+------------+
2 rows in set (0.00 sec)

mysql>
mysql> SELECT *
    -> FROM vw_player_details
    -> LIMIT 10;
+-----------+---------------+-----------+----------------+-----------------------+---------------+
| player_id | player_name   | country   | batting_style  | bowling_style         | date_of_birth |
+-----------+---------------+-----------+----------------+-----------------------+---------------+
|         1 | Virat Kohli   | India     | Right-handed   | NULL                  | NULL          |
|         2 | M Klinger     | Australia | Right hand Bat |                       | 1980-07-04    |
|         3 | TM Head       | Australia | Left hand Bat  | Right arm Offbreak    | 1993-12-29    |
|         4 | MC Henriques  | Australia | Right hand Bat | Right arm Fast medium | 1987-02-01    |
|         5 | AJ Turner     | Australia | Right hand Bat | Right arm Offbreak    | 1993-01-25    |
|         6 | JP Faulkner   | Australia | Right hand Bat | Left arm Fast medium  | 1990-04-29    |
|         7 | TD Paine      | Australia | Right hand Bat | Right arm Medium      | 1984-12-08    |
|         8 | PJ Cummins    | Australia | Right hand Bat | Right arm Fast        | 1993-05-08    |
|         9 | A Zampa       | Australia | Right hand Bat | Legbreak Googly       | 1992-03-31    |
|        10 | B Stanlake    | Australia | Left hand Bat  | Right arm Fast        | 1994-11-04    |
+-----------+---------------+-----------+----------------+-----------------------+---------------+
10 rows in set (0.00 sec)
```

```
mysql>
mysql> SELECT *
    -> FROM vw_batting_performance
    -> LIMIT 10;
+-----------+-----------------+-----------+--------------+----------+--------------+-------------+-------+-------+-------------+------------------+----------+
| player_id | player_name     | country   | innings_id | match_id | runs_scored | balls_faced | fours | sixes | strike_rate | batting_position | phase    |
+-----------+-----------------+-----------+--------------+----------+--------------+-------------+-------+-------+-------------+------------------+----------+
|         1 | Virat Kohli     | India     |            1 |        1 |           43 |          34 |     2 |     2 |      126.47 |                1 | Middle   |
|         2 | M Klinger       | Australia |            1 |        1 |           38 |          32 |     4 |     0 |      118.75 |                2 | Powerplay |
|         3 | TM Head         | Australia |            1 |        1 |           31 |          25 |     0 |     1 |      124.00 |                3 | Middle   |
|         4 | MC Henriques    | Australia |            1 |        1 |           17 |          10 |     0 |     1 |      170.00 |                4 | Middle   |
|         5 | AJ Turner       | Australia |            1 |        1 |           18 |          12 |     1 |     0 |      150.00 |                5 | Middle   |
|         6 | JP Faulkner     | Australia |            1 |        1 |           14 |           7 |     2 |     0 |      200.00 |                6 | Death    |
|        12 | N Dickwella     | Sri Lanka |            2 |        1 |           30 |          25 |     4 |     1 |      120.00 |                1 | Powerplay |
|        13 | WU Tharanga     | Sri Lanka |            2 |        1 |            0 |           2 |     0 |     0 |        0.00 |                2 | Powerplay |
|        14 | EMDY Munaweera  | Sri Lanka |            2 |        1 |           44 |          29 |     6 |     0 |      151.72 |                3 | Powerplay |
|        15 | DAS Gunaratne   | Sri Lanka |            2 |        1 |           62 |          37 |     7 |     0 |      140.54 |                4 | Middle   |
+-----------+-----------------+-----------+--------------+----------+--------------+-------------+-------+-------+-------------+------------------+----------+
10 rows in set (0.00 sec)
```

```
mysql>
mysql> SHOW CREATE VIEW vw_player_details;
+--------------------+--------------------
```

```
mysql>
mysql> CALL GetPlayerBattingStats('Virat Kohli');
+-----------+-------------+---------+------------------+------------+-------------+-------------+
| player_id | player_name | country | innings_recorded | total_runs | total_fours | total_sixes |
+-----------+-------------+---------+------------------+------------+-------------+-------------+
|         1 | Virat Kohli | India   |              242 |       8308 |         822 |         243 |
+-----------+-------------+---------+------------------+------------+-------------+-------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> CALL GetPlayerBattingStats('Unknown Player');
Empty set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

5.12 Trigger

```
mysql> SHOW TRIGGERS;
+-------------------------------+--------+-------------------+-------------------------------
```

```
mysql> SHOW INDEX FROM batting_performance;
+---------------------+------------+-------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table               | Non_unique | Key_name                      | Seq_in_index | Column_name  | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+---------------------+------------+-------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| batting_performance |          0 | PRIMARY                       |            1 | batting_perf_id | A         |      140845 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| batting_performance |          1 | innings_id                    |            1 | innings_id   | A         |       17557 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| batting_performance |          1 | idx_batting_player_innings    |            1 | player_id    | A         |        7060 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| batting_performance |          1 | idx_batting_player_innings    |            2 | innings_id   | A         |      140845 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
+---------------------+------------+-------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
4 rows in set (0.01 sec)

mysql>
mysql> SHOW INDEX FROM bowling_performance;
+---------------------+------------+-------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table               | Non_unique | Key_name                      | Seq_in_index | Column_name  | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+---------------------+------------+-------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| bowling_performance |          0 | PRIMARY                       |            1 | bowling_perf_id | A         |      105447 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| bowling_performance |          1 | innings_id                    |            1 | innings_id   | A         |       17297 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| bowling_performance |          1 | idx_bowling_player_innings    |            1 | player_id    | A         |        6203 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| bowling_performance |          1 | idx_bowling_player_innings    |            2 | innings_id   | A         |      103104 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
+---------------------+------------+-------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
4 rows in set (0.01 sec)

mysql>
mysql> SHOW INDEX FROM `match`;
+-------+------------+-------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table | Non_unique | Key_name                      | Seq_in_index | Column_name  | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-------+------------+-------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| match |          0 | PRIMARY                       |            1 | match_id     | A         |        8796 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| match |          1 | team1_id                      |            1 | team1_id     | A         |         110 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| match |          1 | team2_id                      |            1 | team2_id     | A         |         109 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| match |          1 | weather_id                    |            1 | weather_id   | A         |          96 |     NULL | NULL   | YES  | BTREE      |         |               | YES     | NULL       |
| match |          1 | winner_team_id                |            1 | winner_team_id | A         |         186 |     NULL | NULL   | YES  | BTREE      |         |               | YES     | NULL       |
| match |          1 | idx_match_venue_format        |            1 | venue_id     | A         |         613 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| match |          1 | idx_match_venue_format        |            2 | match_format | A         |         878 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
+-------+------------+-------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
7 rows in set (0.01 sec)

mysql>
mysql> SHOW INDEX FROM opponent;
+----------+------------+-------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table    | Non_unique | Key_name                      | Seq_in_index | Column_name  | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+----------+------------+-------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| opponent |          0 | PRIMARY                       |            1 | opponent_id  | A         |      194848 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| opponent |          0 | player_id                     |            1 | player_id    | A         |        9260 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| opponent |          0 | player_id                     |            2 | team_id      | A         |       55355 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| opponent |          0 | player_id                     |            3 | match_id     | A         |      193804 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| opponent |          1 | team_id                       |            1 | team_id      | A         |         107 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| opponent |          1 | match_id                      |            1 | match_id     | A         |        8861 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| opponent |          1 | idx_opponent_player_team      |            1 | player_id    | A         |        9652 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
| opponent |          1 | idx_opponent_player_team      |            2 | team_id      | A         |       54362 |     NULL | NULL   |      | BTREE      |         |               | YES     | NULL       |
+----------+------------+-------------------------------+--------------+--------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
8 rows in set (0.02 sec)
```

5.14 Transactions and ACID properties

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO player
    -> (player_name, country, batting_style)
    -> VALUES
    -> ('Transaction Demo Player', 'India', 'Right hand bat');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> SELECT *
    -> FROM player
    -> WHERE player_name = 'Transaction Demo Player';
+-----------+---------------------------+---------+-----------------+-----------------+---------------+
| player_id | player_name               | country | batting_style   | bowling_style   | date_of_birth |
+-----------+---------------------------+---------+-----------------+-----------------+---------------+
|      8017 | Transaction Demo Player   | India   | Right hand bat  | NULL            | NULL          |
+-----------+---------------------------+---------+-----------------+-----------------+---------------+
1 row in set (0.00 sec)

mysql>
mysql> UPDATE player
    -> SET country = 'Test Country'
    -> WHERE player_name = 'Transaction Demo Player';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT *
    -> FROM player
    -> WHERE player_name = 'Transaction Demo Player';
+-----------+---------------------------+--------------+-----------------+-----------------+---------------+
| player_id | player_name               | country      | batting_style   | bowling_style   | date_of_birth |
+-----------+---------------------------+--------------+-----------------+-----------------+---------------+
|      8017 | Transaction Demo Player   | Test Country | Right hand bat  | NULL            | NULL          |
+-----------+---------------------------+--------------+-----------------+-----------------+---------------+
1 row in set (0.01 sec)

mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT *
    -> FROM player
    -> WHERE player_name = 'Transaction Demo Player';
Empty set (0.01 sec)
```

```
mysql>
mysql> SELECT *
    -> FROM player
    -> WHERE player_name = 'Transaction Demo Player';
Empty set (0.01 sec)

mysql> WITH PlayerTotals AS (
    ->     SELECT
    ->         p.player_id,
    ->         p.player_name,
    ->         p.country,
    ->         SUM(bp.runs_scored) AS total_runs
    ->     FROM player p
    ->     JOIN batting_performance bp
    ->     ON p.player_id = bp.player_id
    ->     GROUP BY p.player_id, p.player_name, p.country
    -> )
    -> SELECT *
    -> FROM PlayerTotals;
+-----------+------------------------------+--------------------------+------------+
| player_id | player_name                  | country                  | total_runs |
+-----------+------------------------------+--------------------------+------------+
|         1 | Virat Kohli                  | India                    |       8308 |
|         2 | M Klinger                    | Australia                |        143 |
|         3 | TM Head                      | Australia                |       4283 |
|         4 | MC Henriques                 | Australia                |        650 |
|         5 | AJ Turner                    | Australia                |        302 |
|         6 | JP Faulkner                  | Australia                |       1184 |
|        12 | N Dickwella                  | Sri Lanka                |       2084 |
|        13 | WU Tharanga                  | Sri Lanka                |       6889 |
|        14 | EMDY Munaweera               | Sri Lanka                |        230 |
|        15 | DAS Gunaratne                | Sri Lanka                |        800 |
|        16 | TAM Siriwardana              | Sri Lanka                |        791 |
|        17 | CK Kapugedera                | Sri Lanka                |       2180 |
|        18 | S Prasanna                   | Sri Lanka                |        635 |
|        23 | BR Dunk                      | Australia                |         99 |
|         7 | TD Paine                     | Australia                |        943 |
|         8 | PJ Cummins                   | Australia                |        680 |
|        11 | AJ Tye                       | Australia                |        140 |
|        24 | JA Richardson                | Australia                |        138 |
|        25 | BKG Mendis                   | Sri Lanka                |       7001 |
|        21 | KMDN Kulasekara              | Sri Lanka                |       1519 |
|        20 | SL Malinga                   | Sri Lanka                |        644 |
|        26 | MD Shanaka                   | Sri Lanka                |       3133 |
|        19 | JRMVB Sanjaya                | Sri Lanka                |         20 |
|        27 | Nizakat Khan                 | Hong Kong                |       2896 |
|        28 | Aizaz Khan                   | Hong Kong                |       1218 |
|        29 | Babar Hayat                  | Hong Kong                |       2954 |
|        30 | Anshuman Rath                | Hong Kong                |       2700 |
|        31 | Ehsan Khan                   | Hong Kong                |        476 |
|        32 | Tanwir Afzal                 | Hong Kong                |        317 |
|        33 | Shahid Wasif                 | Hong Kong                |        612 |
|        38 | WTS Porterfield              | Ireland                  |       3390 |
|        39 | DA Rankin                    | Ireland                  |         15 |
|        40 | SP Terry                     | Ireland                  |         20 |
|        41 | JNK Shannon                  | Ireland                  |        111 |
|        43 | GJ Thompson                  | Ireland                  |         93 |
|        42 | KJ O'Brien                   | Ireland                  |       3561 |
|        44 | AR McBrine                   | Ireland                  |        946 |
|        45 | L Tucker                     | Ireland                  |       2420 |
|        46 | J Mulder                     | Ireland                  |          1 |
|        47 | J Little                     | Ireland                  |        204 |
|        48 | CA Young                     | Ireland                  |        222 |
|        60 | CJ Chibhabha                 | Zimbabwe                 |       2499 |
|        61 | H Masakadza                  | Zimbabwe                 |       5185 |
|        62 | R Mutumbami                  | Zimbabwe                 |        583 |
|        63 | Sikandar Raza                | Zimbabwe                 |       6392 |
|        64 | MN Waller                    | Zimbabwe                 |       1546 |
|        65 | CT Mutombodzi                | Zimbabwe                 |        235 |
|        66 | E Chigumbura                 | Zimbabwe                 |       4035 |
|        67 | AG Cremer                    | Zimbabwe                 |        572 |
|        68 | N Madziva                    | Zimbabwe                 |        121 |
|        49 | KL Rahul                     | India                    |       5432 |
|        51 | AT Rayudu                    | India                    |       1679 |
|        50 | Mandeep Singh                | India                    |        146 |
```

```
mysql> WITH PlayerTotals AS (
    ->     SELECT
    ->         p.player_id,
    ->         p.player_name,
    ->         SUM(bp.runs_scored) AS total_runs
    ->     FROM player p
    ->     JOIN batting_performance bp
    ->     ON p.player_id = bp.player_id
    ->     GROUP BY p.player_id, p.player_name
    -> )
    -> SELECT
    ->     player_name,
    ->     total_runs,
    ->     RANK() OVER (
    ->         ORDER BY total_runs DESC
    ->     ) AS player_rank
    -> FROM PlayerTotals
    -> ORDER BY player_rank;
+------------------------------+------------+-------------+
| player_name                  | total_runs | player_rank |
+------------------------------+------------+-------------+
| V Kohli                      |      18788 |           1 |
| RG Sharma                    |      15574 |           2 |
| KC Sangakkara                |      13022 |           3 |
| MS Dhoni                     |      11858 |           4 |
| AB de Villiers               |      11026 |           5 |
| TM Dilshan                   |      10959 |           6 |
| Babar Azam                   |      10873 |           7 |
| MJ Guptill                   |      10642 |           8 |
| LRPL Taylor                  |      10100 |           9 |
| Q de Kock                    |      10000 |          10 |
| DA Warner                    |       9877 |          11 |
| JC Buttler                   |       9710 |          12 |
| KS Williamson                |       9682 |          13 |
| DPMD Jayawardene             |       9531 |          14 |
| EJG Morgan                   |       9396 |          15 |
| Tamim Iqbal                  |       9272 |          16 |
| HM Amla                      |       9106 |          17 |
| S Mandhana                   |       8968 |          18 |
| Shakib Al Hasan              |       8933 |          19 |
| Mohammad Hafeez              |       8788 |          20 |
| S Dhawan                     |       8492 |          21 |
| JE Root                      |       8459 |          22 |
| Mushfiqur Rahim              |       8403 |          23 |
| Shoaib Malik                 |       8373 |          24 |
| CH Gayle                     |       8365 |          25 |
| Virat Kohli                  |       8308 |          26 |
| L Wolvaardt                  |       8217 |          27 |
| Yuvraj Singh                 |       8068 |          28 |
| SD Hope                      |       7531 |          29 |
| MJ Clarke                    |       7476 |          30 |
| RT Ponting                   |       7380 |          31 |
| Mahmudullah                  |       7354 |          32 |
| DA Miller                    |       7351 |          33 |
| BB McCullum                  |       7302 |          34 |
| SK Raina                     |       7134 |          35 |
| BKG Mendis                   |       7001 |          36 |
| AD Mathews                   |       6989 |          37 |
| F du Plessis                 |       6932 |          38 |
| BL Mooney                    |       6894 |          39 |
```

6. NoSQL / VECTOR DATABASE COMMANDS

MongoDB is used as the NoSQL database in the Cricket Player Performance & Selection Intelligence System. Unlike relational databases such as MySQL, MongoDB stores information in flexible, document-oriented structures.

In this project, MongoDB is used to store scouting reports and player observations. These documents can contain flexible information related to player evaluation, recommendations and scouting notes.

The MongoDB database is named cricket_selection_nosql, and the collection used for scouting information is named scouting_reports.

This section demonstrates MongoDB database selection, collection inspection, document insertion, retrieval, updating, deletion and aggregation operations.

6.1 Database Selection and Collection Inspection

The following commands are used to select the project database and display the available collections.

MongoDB Commands:

```
>_MONGOSH

> use("cricket_selection_nosql")
< switched to db cricket_selection_nosql
> show dbs

  use cricket_selection_nosql

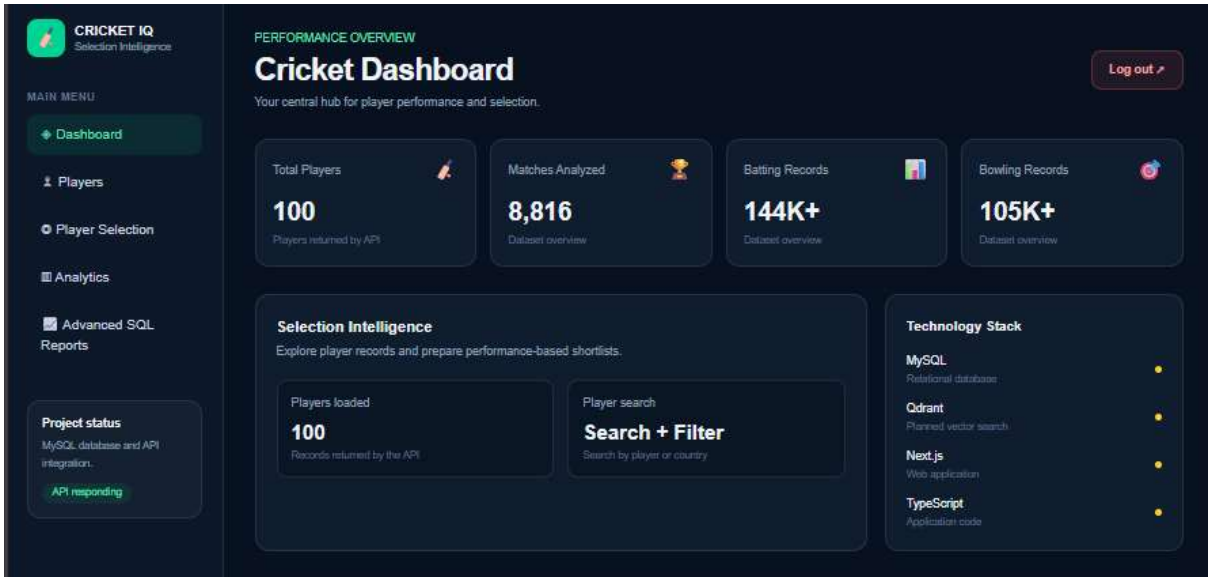
  show collections
< admin                   40.00 KiB
  config                  36.00 KiB
  cricket_selection_nosql 40.00 KiB
  local                   72.00 KiB
cricket_selection_nosql>
```

```
> db.scouting_reports.countDocuments()
< 3
> db.scouting_reports.findOne()
< {
    _id: ObjectId('6ae5a987be17bc508e85a1c4'),
    playerId: 1,
    playerName: 'Virat Kohli',
    recommendation: 'Highly Recommended',
    role: 'Batsman',
    strengths: [
      'Consistency',
      'Chasing ability',
      'Experience'
    ],
    notes: 'Strong batting performance and match experience.'
  }
```

```
> db.scouting_reports.find().limit(5)
< {
    _id: ObjectId('6ae5a987be17bc508e85a1c4'),
    playerId: 1,
    playerName: 'Virat Kohli',
    recommendation: 'Highly Recommended',
    role: 'Batsman',
    strengths: [
      'Consistency',
      'Chasing ability',
      'Experience'
    ],
    notes: 'Strong batting performance and match experience.'
  }
  {
    _id: ObjectId('6ae5a987be17bc508e85a1c5'),
    playerId: 4405,
    playerName: 'A Adekunle',
    recommendation: 'Recommended',
    role: 'Batsman',
    strengths: [
      'Batting technique',
      'Consistency'
    ],
    notes: 'Promising player with good batting potential.'
  }
```

```
  {
    _id: ObjectId('6ae5a987be17bc508e85a1c6'),
    playerId: 2385,
    playerName: 'A Halbhavi',
    recommendation: 'Under Observation',
    role: 'All-rounder',
    strengths: [
      'Versatility',
      'Fielding'
    ],
    notes: 'Requires further performance evaluation.'
  }
```

Section 8 – Code and Demo Screenshots



RULE-BASED ANALYSIS

Player Selection Rankings

Batters ranked using database-derived runs, strike rate and average runs per recorded batting entry.

RANK	PLAYER	COUNTRY	RUNS	STRIKE RATE	SCORE / 100	CATEGORY
1	Karanbir Singh	Austria	2,537	174.72	96.71	High
2	L Barnett	Isle of Man	575	157.53	93.75	High
3	C De Freitas	South Africa	399	171.24	92.93	High
4	Saim Hussain	Bulgaria	577	149.87	90.22	High
5	HAM Shah	Denmark	1,513	134.37	89.55	High
6	D Visser	Samoa	845	142.98	89.55	High
7	Faheem Nazir	Switzerland	775	147.34	89.22	High
8	Sheikh Rasik	Hungary	344	172.86	89.08	High
9	V Kohli	India	18,788	100.74	88.71	High
10	Adil Butt	Saudi Arabia	878	177.02	88.64	High
11	DMW Rawlins	Bermuda	1,235	154.18	88.37	High
12	B Kohler-Cadmore	Germany	434	178.60	88.37	High
13	K Kadowaki-Fleming	Japan	1,779	140.74	88.36	High
14	PV Vuniwaqa	Fiji	533	151.85	88.32	High
15	J Baker	Cayman Islands	1,001	130.17	87.64	High
16	R Pathan	Canada	387	158.61	87.58	High
17	Mohammad Ihsan	Spain	801	176.43	87.43	High
18	Muhammad Fahad	Turkey	380	179.25	87.17	High
19	Taranjeet Singh	Romania	1,644	158.99	87.13	High
20	YBK Jaiswal	India	822	138.38	87.04	High

Prototype ranking for batters only. Scores are rule-based and have not yet been validated as a predictive selection model.

SQL ANALYTICS

Top Batters Report

Ranked by total runs recorded in the dataset.

RANK	PLAYER	COUNTRY	RUNS	RECORDS	FOURS	SIXES	RUNS/RECORD
1	V Kohli	India	18,788	412	1,728	288	45.60
2	RG Sharma	India	15,574	418	1,452	553	37.26
3	KC Sangakkara	Sri Lanka	13,022	337	1,261	97	38.64
4	MS Dhoni	India	11,858	361	901	259	32.85
5	AB de Villiers	South Africa	11,026	285	960	254	38.69
6	TM Dilshan	Sri Lanka	10,959	335	1,232	79	32.71
7	Babar Azam	Pakistan	10,873	268	1,060	146	40.57
8	MJ Guptill	New Zealand	10,642	312	1,041	357	34.11
9	LRPL Taylor	New Zealand	10,100	307	800	207	32.90
10	Q de Kock	South Africa	10,000	263	1,092	259	38.02
11	DA Warner	Australia	9,877	264	1,034	247	37.41
12	JC Buttler	England	9,710	316	833	372	30.73
13	KS Williamson	New Zealand	9,682	254	899	119	38.12
14	DPMD Jayawardene	Sri Lanka	9,531	317	911	85	30.07
15	EJG Morgan	England	9,396	321	785	310	29.27
16	Tamim Iqbal	Bangladesh	9,272	286	1,013	141	32.42
17	HM Amla	South Africa	9,106	217	941	75	41.96
18	S Mandhana	India	8,988	248	1,151	154	36.16
19	Shakib Al Hasan	Bangladesh	8,933	310	854	100	28.82
20	Mohammad Hafeez	Pakistan	8,788	307	881	180	28.63

ADVANCED SQL

Advanced SQL Reports

Top batters ranked using a CTE, JOIN, aggregation, and the RANK window function.

RANK	PLAYER	COUNTRY	TOTAL RUNS
1	V Kohli	India	18,788
2	RG Sharma	India	15,574
3	KC Sangakkara	Sri Lanka	13,022
4	MS Dhoni	India	11,858
5	AB de Villiers	South Africa	11,026
6	TM Dilshan	Sri Lanka	10,959
7	Babar Azam	Pakistan	10,873
8	MJ Guptill	New Zealand	10,642
9	LRPL Taylor	New Zealand	10,100
10	Q de Kock	South Africa	10,000
11	DA Warner	Australia	9,877
12	JC Buttler	England	9,710
13	KS Williamson	New Zealand	9,682
14	DPMD Jayawardene	Sri Lanka	9,531
15	EJG Morgan	England	9,396
16	Tamim Iqbal	Bangladesh	9,272
17	HM Amla	South Africa	9,106
18	S Mandhana	India	8,968
19	Shakib Al Hasan	Bangladesh	8,933
20	Mohammad Hafeez	Pakistan	8,788

Cricket Scouting Reports

Record player observations and recommendations for team selection.

Create Scouting Report

Select Player (from MySQL)

Choose a player

Strengths

Example: Strong batting technique and good timing

Weaknesses

Example: Struggles against short-pitched bowling

Selection Recommendation

Consider

Additional Observations

Optional match observations

Save Scouting Report

Saved Reports

Search by player name...

All Recommendations

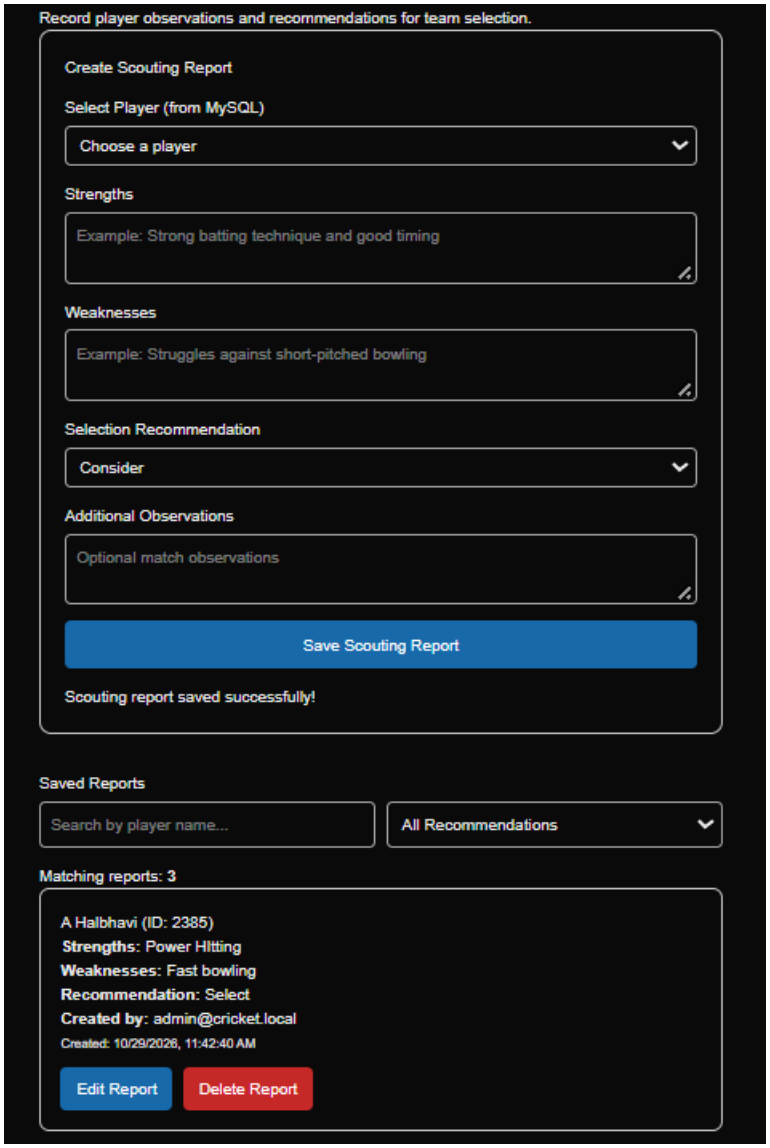
Matching reports: 2

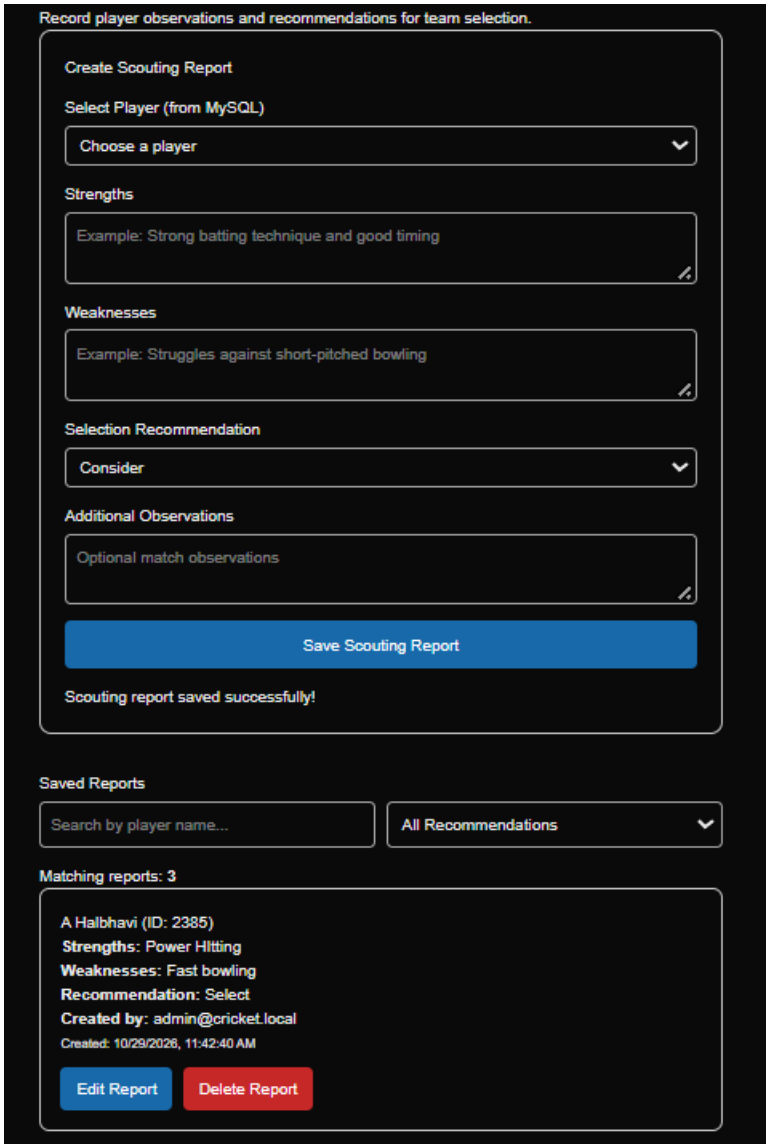
A Adekunle (ID: 4405)
Strengths: hard hitting
Weaknesses: spin bowling
Recommendation: Select
Created by: purimetlasuryantr@gmail.com
Created: 10/28/2026, 6:45:22 PM

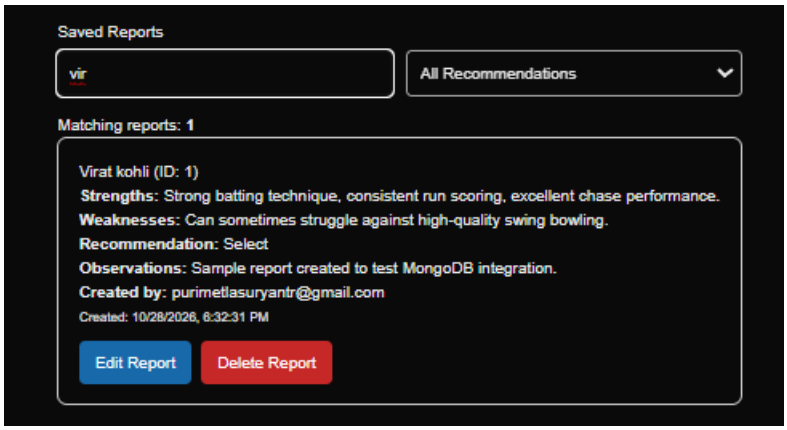
Edit Report Delete Report

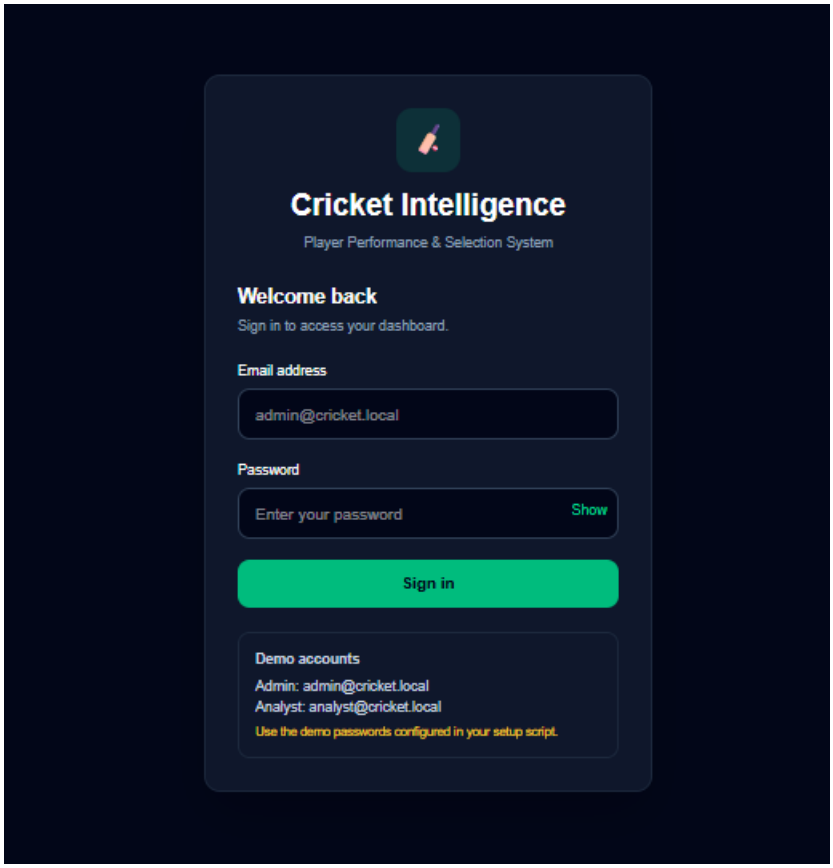
Virat kohli (ID: 1)
Strengths: Strong batting technique, consistent run scoring, excellent chase performance.
Weaknesses: Can sometimes struggle against high-quality swing bowling.
Recommendation: Select
Observations: Sample report created to test MongoDB integration.
Created by: purimetlasuryantr@gmail.com
Created: 10/28/2026, 6:32:31 PM

Edit Report Delete Report


Record player observations and recommendations for team selection.
Create Scouting Report
Select Player (from MySQL)
Choose a player
Strengths
Example: Strong batting technique and good timing
Weaknesses
Example: Struggles against short-pitched bowling
Selection Recommendation
Consider
Additional Observations
Optional match observations
Save Scouting Report
Scouting report saved successfully!
Saved Reports
Search by player name...
All Recommendations
Matching reports: 3
A Halbhavi (ID: 2385)
Strengths: Power Hitting
Weaknesses: Fast bowling
Recommendation: Select
Created by: admin@cricket.local
Created: 10/29/2026, 11:42:40 AM
Edit Report
Delete Report


Record player observations and recommendations for team selection.
Create Scouting Report
Select Player (from MySQL)
Choose a player
Strengths
Example: Strong batting technique and good timing
Weaknesses
Example: Struggles against short-pitched bowling
Selection Recommendation
Consider
Additional Observations
Optional match observations
Save Scouting Report
Scouting report saved successfully!
Saved Reports
Search by player name...
All Recommendations
Matching reports: 3
A Halbhavi (ID: 2385)
Strengths: Power Hitting
Weaknesses: Fast bowling
Recommendation: Select
Created by: admin@cricket.local
Created: 10/29/2026, 11:42:40 AM
Edit Report
Delete Report


Saved Reports
vir
All Recommendations
Matching reports: 1
Virat kohli (ID: 1)
Strengths: Strong batting technique, consistent run scoring, excellent chase performance.
Weaknesses: Can sometimes struggle against high-quality swing bowling.
Recommendation: Select
Observations: Sample report created to test MongoDB integration.
Created by: purimetlasuryantr@gmail.com
Created: 10/28/2026, 6:32:31 PM
Edit Report
Delete Report



9. EVALUATION RUBRIC

The project, **Cricket Player Performance & Selection Intelligence System**, is evaluated based on its database design, implementation, functionality, and documentation.

The following aspects are considered during project evaluation:

1. **Problem Identification:** Identification of a real-world cricket player performance and selection problem.
2. **Database Design:** ER diagram, relational schema, tables, primary keys, and foreign key relationships.
3. **SQL Implementation:** Implementation of SQL queries, joins, subqueries, views, stored procedures, triggers, and other database operations.
4. **NoSQL Implementation:** Integration and demonstration of MongoDB for storing and retrieving scouting-related information.
5. **Application Development:** Integration of the database with the Next.js application.
6. **Functionality and Testing:** Demonstration of implemented features and verification of database operations.

7. **Documentation:** Complete project report, screenshots, code evidence, and demonstration.

The implementation and supporting evidence are presented in the preceding sections of this report.